Study Guide, Midterm 2

ISS 2440: Introduction to Philosophy, Politics, Economics, and Law · Fall 2026

Exam: Monday, October 26, in class. Closed book, closed notes, no devices. Fifteen multiple-choice questions (4 points each) and three short essays, of which you answer two (20 points each). No math and no calculations. Two questions show a game matrix like those in Week 5 below: you find the equilibrium or name the game. Otherwise, questions are on concepts, arguments, and the examples from class and the readings.

What the exam covers: Weeks 5–10 only. Nothing from Weeks 1–4. Not on the exam: Week 9's condominiums, towns, and tribes, and Week 10's lecture on rent seeking.

Required readings: Hobbes, *Leviathan*, ch. 13. Locke, *Second Treatise*, chs. 2–3. Friedman, "A Positive Account of Property Rights" (1994). Olson, "Dictatorship, Democracy, and Development" (1993). Buchanan, "Public Choice: Politics Without Romance" (2003). Tocqueville, *Democracy in America*, vol. I, ch. XV. Caplan, "The Myth of the Rational Voter" (2007).

Week 5. The state of nature (Hobbes, Locke, Rousseau)

- The state of nature is a model of what enforcement is for, not a history. Hobbes: war is a condition, not a battle; men attack for gain (competition), safety (diffidence), and reputation (glory).
- Reading a game. Player 1 (orange) picks a row; Player 2 (blue) picks a column. Each cell gives Player 1's payoff first, then Player 2's; higher is better. An outcome is written (Player 1's choice, Player 2's choice). A *best reply* is the strategy that pays a player most, given what the other does: the arrows point to it. A strategy is *dominant* if it is the best reply whatever the other does. A *Nash equilibrium* (boxed) is an outcome where each player is playing a best reply, so neither gains by changing strategy alone. To test a cell, ask: would Player 1 gain by switching rows? Would Player 2 gain by switching columns? If neither would, the cell is an equilibrium.

Player 1 \ Player 2	Refrain	Attack
Refrain	3, 3 →	0, 5
Attack	5, 0	1, 1

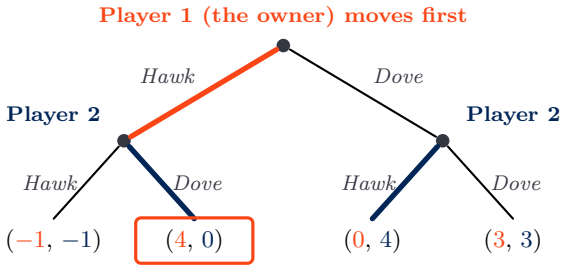
Prisoner's dilemma (Hobbes). Attack is dominant: it pays more whether the other refrains (5 vs. 3) or attacks (1 vs. 0). One equilibrium, (Attack, Attack), although both prefer (Refrain, Refrain).

Player 1 \ Player 2	Stag	Hare
Stag	10, 10	0, 7
Hare	7, 0 →	7, 7

Stag hunt (Rousseau). No dominant strategy: each does best by matching the other. Two equilibria: (Stag, Stag), better for both, and (Hare, Hare), safe. Trust decides which.

Player 1 \ Player 2	Dove	Hawk
Dove	3, 3 →	0, 4
Hawk	4, 0	−1, −1

Hawk–Dove (Locke). No dominant strategy: each does best by doing the opposite of the other. Two equilibria: (Hawk, Dove) and (Dove, Hawk). Someone yields; the game does not say who.



Sequential Hawk–Dove: Locke's point. Same payoffs, but the original owner moves first and the newcomer sees the move before replying. Solve backward. Against Hawk the newcomer yields (0 beats −1); against Dove he takes the land (4 beats 3), as the thick blue lines show. So the owner plays Hawk and keeps the land (4 rather than 0). The owner always defends, the newcomer defers, and no fight happens: possession moves first, and property is a self-enforcing equilibrium. Resources are still lost compared with peace: the pair gets 4, not 6.

- Hobbes: no wickedness is needed for war, only the ranking of outcomes. Leviathan changes the payoffs: once punishment exceeds the temptation, peace is the best reply.
- Repeated play: if people are patient and expect to meet again, the threat to stop cooperating sustains peace without Leviathan (the shadow of the future). A known last round unravels it.
- Commitment: Cortés destroys his ships. Removing your own options can make you better off.
- Locke: a law of nature binds before government; the state of nature has its "inconveniencies" because each man judges his own case.
- Rousseau: the problem is trust, not temptation. Conventions such as Hume's two rowers pick the good equilibrium.
- Three solutions: Hobbes, a sovereign who changes the payoffs; Locke, limited government that enforces rights already there; Rousseau, a general will that selects the good equilibrium.

Week 6. Is anarchy so bad? (Friedman, Leeson)

- Anarchy is the absence of a state, not the absence of order. Weber: the state claims "the monopoly of the legitimate use of physical force"; Friedman: if "legitimate" means legal, the definition is circular.
- Friedman: rights exist before the state as Schelling (focal) points, defended by a commitment to fight even small violations (the Falklands). "The laws and customs of civil society are an elaborate network of Schelling points."
- Norms first, law later. Whalers: "fast-fish, loose-fish" for slow right whales, "iron holds the whale" for sperm whales that swim in schools. Ellickson: close-knit groups adopt norms that maximize their wealth; judges deferred to them. The limit: overwhaling. No state court enforced their deals, yet merchants (the Law Merchant) and brokers expelled cheats: multilateral reputation is the shadow of the future with everyone as the punisher.
- Unequal strength: Leeson's armed caravans traded with unarmed villages on credit. A village in debt has nothing to steal and is an asset the caravan protects.
- Anarchy compared to what? The nirvana fallacy (judging a real institution against an ideal one) cuts both ways. A state pays only if it adds more wealth than it costs: little in small, close-knit groups, too much for a world state. Leeson's relevant alternative: compare a stateless society with the government it actually had. Somalia: after Siad Barre's predatory state collapsed in 1991, most indicators improved, by more than in neighboring countries; attempts to restore a state brought violence back. None of this says the United States would do better without a state.

Week 7. The origins of the state (Olson, Mayshar–Moav–Pascali, Buchanan)

- Olson: the roving bandit takes everything and leaves; the stationary bandit takes a regular share, a tax, and keeps rivals out, so his victims prefer him (1920s China). He is "led, as though by an invisible hand, to settle down, wear a crown"; nobody signs a contract, because in a large group no one would pay to organize one.
- Encompassing interest: the stationary bandit gains when his subjects do, so he builds roads, though too few. Horizon: a ruler who expects to fall soon turns roving bandit; hence "long live the king." A democratic majority also earns market income, so it taxes less than an autocrat.
- The surplus story (Diamond) predicts states on the richest land. But better farming buys more people, not a surplus (Malthus), and an elite that seizes grain creates the surplus it eats.
- Appropriability (Mayshar, Moav, Pascali): cereals are harvested at once, stored, and easy to find and seize; roots and tubers stay in the ground. Prediction: states where land favors cereals, however rich it is (Egypt vs. New Guinea). "History records no cassava states" (Scott). A rival theory (Tilly): war made the state.
- Buchanan: the social contract is a test (could everyone agree to these rules?), not a history. Some contracts were signed: the Mayflower Compact, pirate articles. Locke's tacit consent has critics: Hume's passenger, carried aboard asleep, does not consent by staying; Spooner: even if your grandparents signed, you did not. In anarchy each side arms against the other; mutual disarmament is "the first leap out of the anarchistic jungle." Olson and Buchanan end at the same question: who restrains the enforcer?

Week 8. Politics without romance (Buchanan, Rousseau, Condorcet)

- Romance treats "the government" as a benevolent being. Test: would I back this policy if my worst opponent ran it?
- Public choice applies economics to politics. Behavioral symmetry: the same people, with the same motives, in markets and in politics; nobody turns into one of Buchanan's "economic eunuchs" in the voting booth. Market failure must be compared with government failure, not with an ideal government.
- Rousseau: the general will (the common good) is not the will of all (a sum of private wills), and it is "found by counting votes." Robespierre saw that in rulers "every will seeks to dominate," but not in the people; once a majority can be called misled, whoever claims to know the general will can overrule it.
- Condorcet's paradox: three voters with consistent rankings, and a majority ranking that goes in a circle. Whoever sets the agenda picks the winner. Buchanan: cycles protect minorities, because majorities rotate.
- Arrow: with three or more options, no voting rule always turns individual rankings into a consistent group ranking while meeting a few mild conditions, unless one voter dictates. Groups need not cycle; no rule guarantees they won't. Counting votes does not find the general will; the rule and the agenda make one.
- The paradox of voting: your vote decides only if the others tie (in 2008, about a 1 in 60 million chance), yet about 64% voted in 2024. Why: altruism (cheap in the booth, but so are ignorance and anger), duty (circular if it only restates that people vote), social pressure (in Michigan, being seen raised turnout more than a reminder of duty), and expressive voting (fans who say "we won").
- Rational ignorance (Downs): weeks on a phone, minutes on a candidate, because one vote almost never decides.

Week 9. Constitutional choice (Tocqueville, Buchanan–Tullock)

- The pizza vote: 24 students can make 6 pay, which is unfair and wasteful. Choosing a rule before you know your role is a different choice from voting on one proposal.

- Tocqueville: a majority is an interested party, and "men are not apt to change their characters by agglomeration": the tyranny of the majority. His target is unlimited power, whoever holds it.
- Unanimity is the benchmark (a voluntary trade is unanimity between two); its price is the holdout. Juries use it; Poland's liberum veto shows the danger: a veto anyone holds is a veto anyone can buy.
- Buchanan and Tullock: as the share who must agree rises, decision-making costs rise and external costs (what others can do to you) fall. The best rule minimizes their sum, and nothing puts it at 50% plus one. Higher stakes push toward unanimity (juries, amendments); urgency pushes toward fewer deciders.
- Hayek: planning needs billions of decisions, so assemblies become "talking shops" and power passes to experts; capitalism is necessary for political freedom, not sufficient. In a bigger House each member counts less; power drifts to leaders (Madison: "more democratic, but ... more oligarchic"). Small boards (Jensen) with one goal decide easily.

Week 10. The median voter and interest groups (Caplan, Olson, Stigler)

- Hotelling: stores and parties crowd to the middle of Main Street (Wallace: "not a dime's worth of difference").
- Median voter theorem: with one issue, single-peaked preferences, two candidates, and everyone voting, both candidates converge on the median voter's favorite, which beats every alternative. Sounding alike is a sign of competition. Single peaks also rule out Week 8's cycles.
- Meltzer–Richard: median income is below the mean, so the median voter gains from some redistribution; he stops where the work it discourages costs him more than the check brings.
- Majority rule counts heads, not intensity: a footbridge worth more than it costs loses 6 to 4.
- Caplan: random errors cancel (the miracle of aggregation); systematic errors do not. Four biases: anti-market, anti-foreign, make-work, pessimistic. Rational irrationality: a wrong belief costs the voter nothing in the booth, so he keeps the one that feels good, and competition delivers the median voter's mistakes.
- Olson against Marx: a shared interest is not enough, because each member of a large group free-rides; small groups organize, and large ones need selective incentives, benefits for members only (the AMA, the Farm Bureau). Concentrated benefits, dispersed costs: the sugar program costs each consumer a few dollars and is worth six figures to the average sugar farm; it is why American Coke uses corn syrup. Stigler: regulation "is acquired by the industry."